The Black-Scholes equation is written with Greeks

The hedge from #6 leaves three derivatives of the option price. Each one has a name.

$$\frac{\partial f}{\partial t} + rS\frac{\partial f}{\partial S} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 f}{\partial S^2} = rf$$

$$\Theta + rS\,\Delta + \frac{1}{2}\sigma^2 S^2\,\Gamma = rf$$

Theta $\Theta = \partial f/\partial t$

Time decay: the value lost as time passes, everything else equal.

Delta $\Delta = \partial f/\partial S$

Slope of the price curve: shares held per option in the hedge.

Gamma $\Gamma = \partial^2 f/\partial S^2$

Curvature of the price curve: how fast delta moves, so how often to rehedge.

Vega ($\partial f/\partial \sigma$) and rho ($\partial f/\partial r$) are not in the equation: σ and r are its constants.

The five Greeks of a call

S = K = 100, r = 5%, vol = 20%, T = 1y, no dividends. Shaded: the three Greeks in the equation.

Greek	What it measures	Formula (call)	Value	Desk unit
Delta $\partial f/\partial S$	Price change per $1 move. Shares held per option in the hedge.	$N(d_1)$	**0.637**	per $1 move
Gamma $\partial^2 f/\partial S^2$	Change in delta per $1 move. Curvature of the price.	$\frac{N'(d_1)}{S\sigma\sqrt{T}}$	**0.0188**	per $1 move
Theta $\partial f/\partial t$	Price change as time passes.	$-\frac{S N'(d_1)\sigma}{2\sqrt{T}} - rKe^{-rT}N(d_2)$	**-6.41 /yr**	-0.018 per day
Vega $\partial f/\partial \sigma$	Price change per vol move. Not in the equation.	$S\sqrt{T}\,N'(d_1)$	**37.5**	0.375 per vol pt
Rho $\partial f/\partial r$	Price change per rate move.	$KTe^{-rT}N(d_2)$	**53.2**	0.532 per 1% rate

$d_1 = \frac{\ln(S/K) + (r + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}$, $d_2 = d_1 - \sigma\sqrt{T}$, N = standard normal CDF, N' = its density.

What it says about a hedged option

Reading the equation: set delta to zero, then set the rate to zero.

Delta-neutral

$$\Theta + \tfrac{1}{2}\sigma^2 S^2 \Gamma = rf$$

A hedged option earns the riskless rate. Time decay is what you pay for the gamma you hold.

And if r = 0

$$\Theta = -\tfrac{1}{2}\sigma^2 S^2 \Gamma$$

An exact exchange rate: what time takes, gamma gives back when realised vol equals the model's vol. Post #9.

Vega and rho

$$\sigma,\ r \text{ constant}$$

They are not results of the derivation: traders bump vol and rate in the model. A stochastic vol model would be the consistent way.

Check on the call above: -6.41 + 3.18 + 3.75 = 0.5225 = 5% x 10.4506.